Regularization

Regularization: Techniques to increase the model bias.

a) parameter penalty

b) Architectural restrictions

c) optimization procedures.

Regularization via parameter penalty

Recall, ERM:

$$\hat{h}_{\hat{\theta}}(x) = \underset{h \in \mathcal{H}}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^{N} L\left(h_\theta(x_i), y_i\right)$$

Regularized ERM:

Define a function $\Omega(\theta) : \Theta \to \mathbb{R}^+$

↳ regularizer

& define a new-objective as follows:

Reg-ERM:

$$\hat{h}^*_\theta(x) = \underset{h_\theta}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^{N} L\left(h_\theta(x_i), y_i\right)$$

subject to $\Omega(\theta) < K$.

where $K \in \mathbb{R}^+$

To solve reg-ERM:

$$\hat{R}_r(\theta) = \frac{1}{N} \sum_{i=1}^{N} L\left(h_\theta(x_i), y_i\right) + \lambda \left(\Omega(\theta) - K\right)$$

reg. const.

$h_\theta^* = \underset{\theta}{\operatorname{argmin}}\ \hat{R}_r(\theta)$, solve analytically or numerically.

Choices for $\Omega(\theta)$

i) $\Omega(\theta) = \|\theta\|_p$

$\|\theta\|_2$: L_2-reg | Ridge.

$\|\theta\|_1$: L_1-reg | Lasso

ii) $\Omega(\theta) = \exp(-\theta)$

Equivalence b/w Reg-ERM & MAP Estimates

Recall,

In Map-estimate we seek;

$$p(\theta \mid x, y) \propto p(y \mid x, \theta) \cdot p(\theta)$$

consider an Example where

$$p(y \mid x, \theta) \sim \mathcal{N}\left(y ; h_\theta(x), I\right)$$

$$l(\theta) = \frac{1}{N} \sum_{i=1}^{N} \log p(y_i \mid x, \theta)$$

$$\propto \frac{-1}{N} \sum_{i=1}^{N} \| y_i - h_\theta(x_i) \|_2^2$$

suppose $p(\theta) = \mathcal{N}\left(\theta ; 0, I\right)$

$$p(\theta \mid y, x) \propto \underbrace{p(y \mid x, \theta) \cdot p(\theta)}$$

$$l_r(\theta) \propto \log p(\theta \mid y, x)$$

$$\propto \log p(y|x,\theta) + \log p(\theta)$$

$$\propto -\frac{1}{N} \|y_i - h_\theta(x_i)\|_2^2 - \|\theta\|_2^2$$

In prob. treatment,

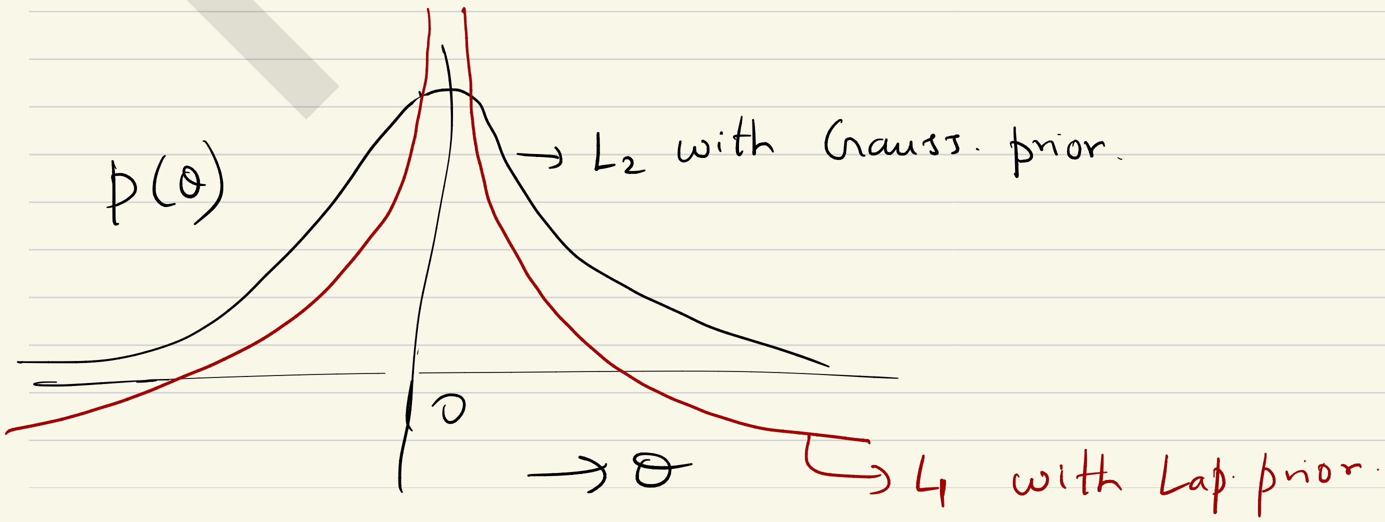
$$\theta^* = \underset{\theta}{\operatorname{argmax}} \; l_r(\theta)$$

$$= \underset{\theta}{\operatorname{argmin}} \; -l_r(\theta)$$

$$= \underset{\theta}{\operatorname{argmin}} \; \frac{1}{N}\sum_{i=1}^{N} \|y_i - h_\theta(x_i)\|_2^2 + \|\theta\|_2^2$$

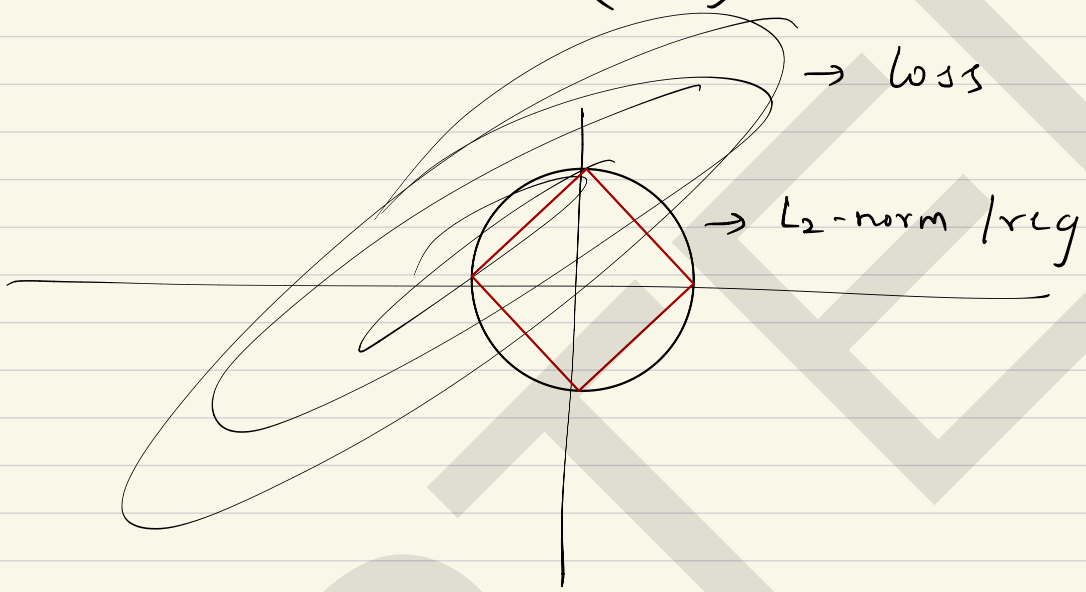
$$= \hat{R}_r(\theta), \text{ with } \Omega(\theta) = \underline{\|\theta\|_2^2}$$

$\Rightarrow$ Reg. ERM with $\|\theta\|_2$ as regularizer is Bayesian estimate with a Gaussian prior.



Suppose $p(\theta) \sim \text{Laplacian}(\alpha)$, then the prob. of parameters around mean will be higher than that of Gaussian.

$$\exp(-|\theta|)$$



Another Example for Regularization.

Suppose $D = \{(x_i, y_i)\} \sim \text{iid } \mathbb{P}_{xy}$

Create a new-dataset as follows,

$\forall x_i \in D$,

$$\hat{x}_i = x_i + \epsilon \qquad \epsilon \sim N(0, \alpha I)$$

$$\hat{D} = \{(\hat{x}_i, y_i)\}$$

<u>Claim:</u> ERM on $\tilde{D}$ $\equiv$ R-ERM on D.

Revisiting Gradient Descent

$$\hat{R}(\theta) = \frac{1}{N} \sum_{i=1}^{N} L\left(h_\theta(x_i), y_i\right)$$

Grad- descent :

$$\theta^{t+1} \leftarrow \theta^t - \beta \cdot \nabla_\theta \hat{R}(\theta)$$

Alter the Grad-descent :

Define $\hat{R}_B(\theta) = \frac{1}{B} \sum_{i=1}^{B} L\left(h_\theta(x_i), y_i\right)$

where $B << N$

& $x_i \in D_B$ where

$D_B \subset$ B-length subset of D.

$$\theta^{t+1} \leftarrow \theta^t - \beta \cdot \nabla_\theta \hat{R}_B(\theta).$$

Stochastic Grad. descent (SGD)